

International AS and A-level

Physics (9630)

PH03 Fields and their consequences

Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL A-LEVEL PHYSICS (9630) PH03 - JANUARY 2025

This paper was similar to those of previous years in terms of specification coverage. Students had many opportunities to demonstrate their knowledge, skills and understanding across a range of topics. The usual variety of question styles including short answers, single and multi-step calculations, extended writing and multiple choice was used. The questions were set in a range of both familiar and unfamiliar contexts.

The balance of assessment objectives and the mathematical demand was similar to those of previous series.

In general, students found this paper more accessible than those of recent series.

QUESTION 01

Question 1 assessed the topics of circular motion and shm in the context of Foucault's pendulum. The majority of students performed well on all of the question parts. The question discriminated well.

Most of the students achieved both marks for question 01.1. A common error that was seen was to use $\sin$ of the angle, but then failing to double the distance. Errors were also seen when converting the angle to radians by some students who chose that route.

The calculation in 01.2 was only slightly more demanding. The majority of students chose to work out the period and use $A\omega$, rather than using a calculation based on energy. Some students tried to base their answer on circular motion and therefore made little progress.

Students who had no problem with question 01.3 calculated the circumference of the path of **P** and divided it by the number of seconds in one day. Several incorrect answers were seen that attempted to calculate the orbital speed of **P**, not taking in the information that the pendulum was on the surface of the Earth. More care when reading the question would have helped avoid this error.

Question 01.4 was the hardest part in question 1, although the majority of students still achieved full marks. For those students who could not complete the calculation, the mark scheme allowed credit for partial answers. Marks could only be awarded if it was clear what the student was attempting to do and therefore students should be encouraged to set out their work clearly at all times.

QUESTION 02

Question 2 tested understanding of gravitation and orbits in the context of a solar satellite. The first three question parts were answered well by most students and discriminated well. The fourth part was more demanding, and discriminated less well.

In question 02.1, students were expected to obtain the mass of the Sun from the data booklet and choose the correct distance. Most students were able to do this, and went on to complete the calculation correctly. Those who chose the incorrect distance were still able to score one mark, provided no other error was made. Some students included the radius of the Sun in their calculation, even though the question gave the distances from the centre.

Question 02.2 was only slightly more demanding than question 02.1. The most common error was made by students who tried to use equations related to circular motion. Partial credit could still be awarded depending on what other steps the students made. The best answers were set out clearly, taking care with the sign of the potential energy.

Question 02.3 discriminated very well. Most students obtained all three marks. The mark scheme was designed to reward students for making some progress. In multi-step calculations like this, students should be encouraged to attempt the question even if they are not sure how to complete it.

Many students were able to identify that C was on an equipotential, but were unable to explain their answer. It was common to see the simple statement that its orbit was circular, without any attempt to link it to the question.

QUESTION 03

Question 3 tested students' understanding of electric fields in the context of the determination of the specific charge of the electron.

Question 03.1 discriminated very well. Students unable to answer the question generally picked up one mark for the use of the kinetic energy formula. There was evidence of some confusion with v (velocity) and V (pd) in several answers. Students should be careful when using similar symbols in expressions.

Most students got at least one mark in Question 03.2, but very few were able to get all three marks. Too many simply restated the information given in the question. The best answers made it clear that the acceleration was constant, explaining it in terms of the uniform field producing a constant force. They then went on to state that the acceleration was downwards, explaining this in terms of the direction of the field and the charge on the electron, and therefore the direction of the force.

It is clear from answers to 03.3 that students are more aware of the expectations of 'show that' questions. Answers were mostly given to more than one significant figure, and evidence was seen of the correct use of powers of ten.

The mark scheme to question 03.4 was designed to reward partially correct answers. Just under half of the students got full marks, and the remainder were almost evenly spread amongst the other marks. This was the most discriminating question on the paper.

Question 03.5 was much more demanding. A wide range of answers was seen and it was clear that many students understood that the electrons would be slowed down and scattered. Descriptions of how this affected what was seen on the screen were less clear.

QUESTION 04

Question 4 tested students' understanding of capacitor charging using a data logger. Students who understood that the points on the graph were plotted from right to left made more progress.

Question 04.1 required students to extrapolate the graph and read off the value on the voltage axis. Most students completed this successfully, although the occasional mis-read was seen.

The most common error in 04.2 was made by students who got the wrong power of ten for the answer, missing the fact that the current was in milliamps. The majority of students obtained full marks for this question, however.

Very few students got the mark for question 04.3. Many students simply stated that it was due to the current getting smaller, without any link to charge or the charging equation.

The multi-step calculation in question 04.4 produced a spread of marks and discriminated well. Several alternative routes were seen. Full credit was given to answers that used a half-life approach as the read-off at 3.1 V could be seen. Attempts to use an e^{-1} approach were less successful as no corresponding point was plotted, and students had to make a guess at the time. The most common problem students had was determining the length of time. As the capacitor was charging, the time was counted from the point on the current axis. Answers that tried to count from the voltage axis gained little credit. Several students also tried to use the discharging equation for voltage.

Question 04.5 spread the students well, with about one third of the students getting each of zero, 1 mark and 2 marks. The most common error by students who got one mark was to continue the line to the edge of the graph paper, rather than stop it when the voltage equalled the emf.

Question 04.6 was much more demanding, with students failing to see that any attempt to record at 5 ms intervals manually was impossible.

QUESTION 05

Question 5 tested students' understanding of radioactive decay in terms of a school-based source.

The 'cognitive step' mark scheme used with question 05.1 discriminated very well. Most students were able to determine the decay constant for the source. Some students had difficulty converting their number of nuclei into a mass. It was clear that some students had little idea of the 'mole' and Avogadro's number in this context.

The calculation in question 05.2 was answered well by the majority of students. There was some evidence of students mixing up the two activities and trying to use negative logs. Most students had little difficulty, however.

QUESTION 06

Question 6 tested students' understanding of magnetism in the context of a demonstration of the motor effect.

There are several definitions in the specification including the tesla, which was tested in question 06.1. There are several points to the definition and few students managed to state all of them. Students who attempted an answer in terms of force often missed out the fact that the current is at right angles to the magnetic field.

Most students got one mark in question 06.2 for correctly identifying which terminal was positive. Far fewer students were able to explain their answer in terms of Fleming's left hand rule and the direction of current.

The mark scheme for question 06.3 meant that it discriminated very well. Most students completed the calculation successfully. Students who attempted to work out the forces vertically made little progress.

The ideas behind question 06.4 have been tested many times before. Despite this, very few fully correct answers were seen. Many students realised the emf would increase but could not explain why. The most common single mark awarded was for the expression of Faraday's law with a reference to a change of flux.

QUESTION 07

Question 7 tested students' knowledge and understanding of oscilloscopes and transformers.

Question 07.1 discriminated very well and produced a good spread of marks. The most successful students set out their answers clearly, showing how they arrived at their settings. They went on to write their answers using a sensibly rounded value. Students who gave 4.99 as the y-gain setting did not get the fourth mark.

The most common problem with question 07.2 was a power of ten error. Most students answered this question successfully, however.

Many of the answers to question 07.3 showed some knowledge of the experiment, but a failure to read the question. Many students simply wrote down what they had apparently learned from previous similar questions.

Most students were able to identify one source of energy loss in a transformer and therefore gain one mark for question 07.4. Those who got two marks made it clear where heat losses were occurring, for example. A large number of students identified hysteresis losses, which was therefore included on the mark scheme, despite this not being on the specification.

SECTION B

The majority of students scored highly on the multiple choice questions.

The easiest question was B9, a calculation of angular speed.

The most demanding questions were B22 and B20. The most popular distractor in B22 was A, chosen by students who believed the emf and flux linkage are in phase. The most popular distractor in B20 was D, chosen by students who did not realise that no work is done on the particle. Question B15 also had a relatively popular distractor: C.